

TSOGT-OVOO, Mongolia

WMO: 443470

Lat: 44.4232N

Lon: 105.3180E

Elev: 4259

StdP: 12.57

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-12.7	-24.3	1.7	-11.2	-20.7	2.1	-8.6	32.4	14.5	30.8	16.4	3.0	320	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	92.0	61.9	88.8	60.6	85.9	59.9	71.7	77.6	67.3	77.2	64.0	77.9	10.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.1	130.1	74.7	64.5	106.8	71.0	59.8	90.0	68.6	38.8	77.3	34.5	76.6	31.7	78.1	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.3	27.6	23.2	-20.7	96.9	5.0	3.4	-24.3	99.3	-27.3	101.3	-30.1	103.2	-33.7	105.7		
			-21.3	69.1	4.7	5.5	-24.7	73.0	-27.5	76.2	-30.1	79.3	-33.6	83.2		
			DB													
			WB													

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	4.4	12.6	28.3	45.2	56.9	68.4	74.0	70.2	58.5	41.7	23.2	8.5
(7)		DBStd	25.64	9.03	10.39	10.92	9.56	8.97	6.97	6.07	6.30	8.34	9.20	10.30	8.70
(8)		HDD50	5772	1413	1048	675	202	38	0	0	0	24	282	805	1285
(9)		HDD65	9381	1878	1468	1139	596	276	44	5	24	224	723	1254	1750
(10)		CDD50	2532	0	0	1	56	253	551	742	627	278	24	0	0
(11)		CDD65	669	0	0	0	1	26	146	283	185	28	0	0	0
(12)		CDH74	6945	0	0	0	15	389	1588	2974	1695	282	2	0	0
(13)		CDH80	2516	0	0	0	1	93	554	1249	572	47	0	0	0
(14)	Wind	WSAvg	9.6	7.6	8.2	10.8	12.3	12.5	10.4	9.2	8.9	8.9	9.1	9.5	7.9
(15)	Precipitation	PrecAvg	3.7	0.0	0.0	0.1	0.1	0.3	0.6	1.2	0.9	0.4	0.2	0.1	0.0
(16)		PrecMax	8.3	0.1	0.4	0.3	0.9	1.7	1.8	3.5	3.5	2.0	0.7	0.3	0.4
(17)		PrecMin	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
(18)		PrecStd	1.5	0.0	0.1	0.1	0.2	0.3	0.4	1.0	0.7	0.4	0.2	0.1	0.1
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	43.4	61.4	76.7	87.4	93.5	98.4	93.6	84.8	71.3	53.4	34.0
(20)			MCWB	24.6	31.3	41.7	51.2	55.9	60.9	67.1	61.8	59.1	49.6	40.4	26.3
(21)		2%	DB	26.1	37.7	56.0	71.7	82.7	89.5	93.5	89.2	80.7	65.5	47.4	28.9
(22)			MCWB	21.0	28.8	39.3	48.5	55.3	59.7	63.0	60.6	55.8	46.1	35.8	23.0
(23)		5%	DB	21.8	33.4	50.7	67.2	78.7	86.0	90.0	86.1	77.1	61.2	42.9	24.9
(24)			MCWB	17.5	25.9	36.2	46.0	52.7	58.8	61.6	60.7	53.9	44.0	33.3	20.0
(25)		10%	DB	18.1	29.1	45.8	62.6	74.4	82.7	87.0	82.8	73.3	56.9	38.7	21.4
(26)			MCWB	14.7	23.2	33.5	43.9	51.3	56.8	60.9	58.9	52.6	41.9	31.0	17.5
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	33.6	46.1	57.1	68.5	77.6	78.6	75.7	66.3	52.8	42.4	26.9
(28)			MCDWB	30.8	40.6	54.0	65.3	73.8	83.1	82.4	80.7	75.2	65.1	48.8	32.8
(29)		2%	WB	21.2	29.5	40.4	51.1	61.4	70.1	72.0	68.8	60.0	47.8	37.6	23.3
(30)			MCDWB	26.0	36.8	53.2	65.4	72.9	74.1	80.0	76.1	72.3	61.6	46.3	28.5
(31)		5%	WB	17.8	26.4	36.8	47.4	55.9	63.4	67.4	64.9	56.6	45.1	34.1	20.4
(32)			MCDWB	21.8	33.2	50.4	64.8	72.3	75.3	78.9	77.5	71.4	59.5	43.0	24.7
(33)		10%	WB	14.8	23.4	33.7	44.4	52.5	60.0	64.3	62.3	53.8	42.6	31.4	17.6
(34)			MCDWB	18.3	29.1	45.6	61.7	72.0	76.6	79.6	76.3	69.9	55.9	39.5	21.3
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.0	22.3	23.0	22.9	22.9	21.2	19.9	19.5	21.0	21.3	19.6	18.3
(36)			MCDBR	22.1	27.0	27.7	27.8	27.4	24.5	23.2	22.9	24.5	26.2	23.8	21.2
(37)			MCWBR	17.9	19.9	17.1	15.4	13.7	10.5	8.5	8.9	11.3	14.6	15.7	16.7
(38)		5% WB	MCDBR	21.8	26.2	26.8	26.6	25.3	21.3	19.7	19.3	22.1	24.1	22.6	20.9
(39)			MCWBR	17.9	19.8	18.0	17.1	16.9	14.1	11.3	10.7	12.6	14.8	16.0	16.6
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.286	0.328	0.373	0.393	0.412	0.393	0.375	0.340	0.305	0.273	0.257	
(41)		taud	2.327	2.265	2.178	2.102	2.096	2.126	2.247	2.275	2.321	2.352	2.354	2.325	
(42)		Ebn at Noon	276	286	286	281	278	272	276	278	279	276	268	264	
(43)		Edh at Noon	25	32	40	48	49	48	42	40	35	29	24	23	
(44)	All-Sky Solar Radiation	RadAvg	664	1007	1536	1952	2251	2225	2109	1934	1630	1209	784	576	
(45)		RadStd	88	125	59	85	108	113	95	78	59	42	65	64	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-3.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-3.98	N/A	-2.12	-2.18	N/A	N/A	N/A	N/A

Nomenclature: See separate page